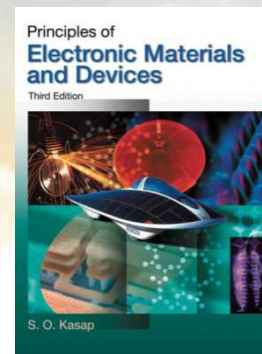




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# 《固态电子学》 Solid-State Electronics



## Ch1-2 Elementary Materials Science Concepts

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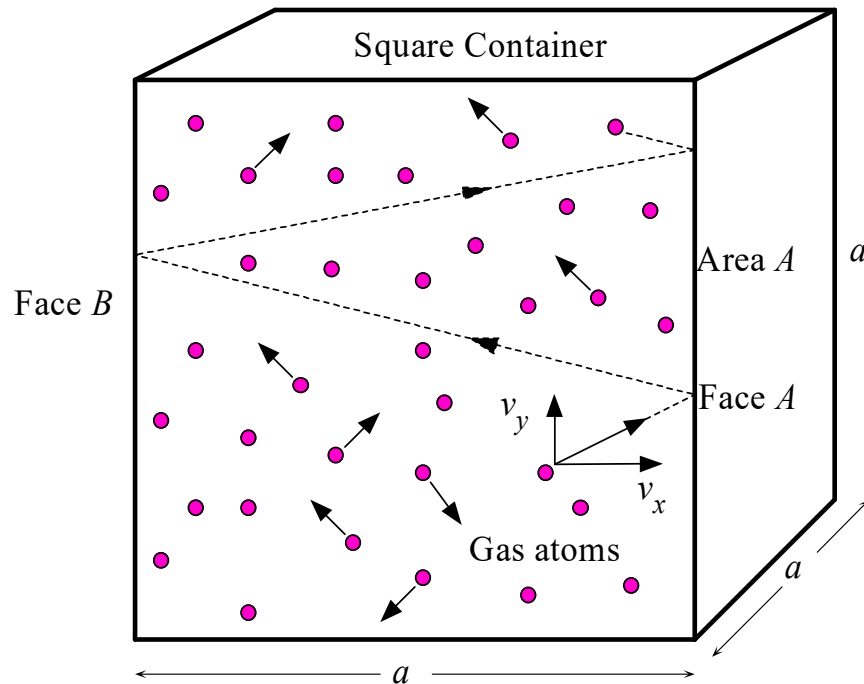
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Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

# 1.4 Kinetic molecular theory 分子动力学理论



Important for

- Pressure of gas
- Heat capacity of metal
- Average speed of electron in semiconductor
- Electrical noise in resistor

The gas molecules in the container are in random motion

Empirical gas equation obtained  
from gas experiments  
气体实验得到的  
经验气体方程

$$PV = \left( \frac{N}{N_A} \right) RT \quad \Delta p = 2m v_x \quad \Delta t = \frac{2a}{v_x}$$

Average kinetic energy per atom  
单个原子的平均动能

$$\overline{KE} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$



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$$\Delta p = 2mv_x \quad \Delta t = \frac{2a}{v_x} \quad \Rightarrow \quad F = \frac{\Delta p}{\Delta t} = \frac{2mv_x}{(2a/v_x)} = \frac{mv_x^2}{a}$$

$$P = \frac{\text{Total force}}{a^2}$$

$$= \frac{mv_{x1}^2 + mv_{x2}^2 + mv_{x3}^2 + \dots + mv_{xN}^2}{a^3}$$

$$= \frac{mN\overline{v_x^2}}{V}$$

$$\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2} \Rightarrow \overline{v^2} = 3\overline{v_x^2}$$

$$PV = \frac{2N}{3} \left( \frac{1}{2} m \overline{v^2} \right)$$

$$\Rightarrow PV = \frac{1}{3} Nm \overline{v^2}$$

$$P = \frac{Nm \overline{v^2}}{3V} = \frac{1}{3} \rho \overline{v^2}$$

Gas density  
气体密度

$$\Rightarrow PV = \left( \frac{N}{N_A} \right) RT = NkT$$

$$\overline{KE} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$

The average kinetic energy of each molecule is proportional to temperature  
每个分子的平均动能与温度成正比

$k=R/N_A$ : Boltzmann constant 波尔兹曼常数  
 $R$ : the universal gas constant 普适气体常数  
 $N_A$ : Avogadro's number 阿伏伽德罗常数



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## Gas (单原子气体)

$$U = N_A \left( \frac{1}{2} m \overline{v^2} \right) = \frac{3}{2} N_A k T$$

Heat capacity 热容: the rise in the internal energy per unit temperature

固定体积内

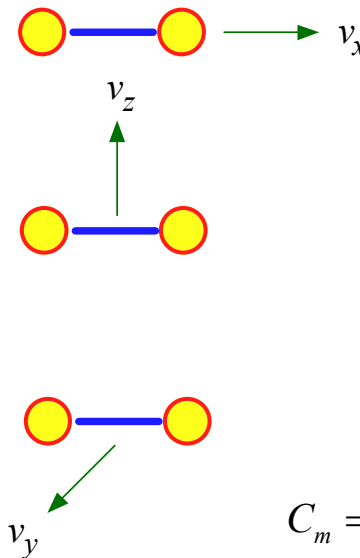
Molar heat capacity 摩尔热容: The heat capacity of one mole gas.

$$C_m = \frac{dU}{dT} = \frac{3}{2} N_A k = \frac{3}{2} R$$

Maxwell's principle of equipartition of energy  
麦克斯韦能量均分定理

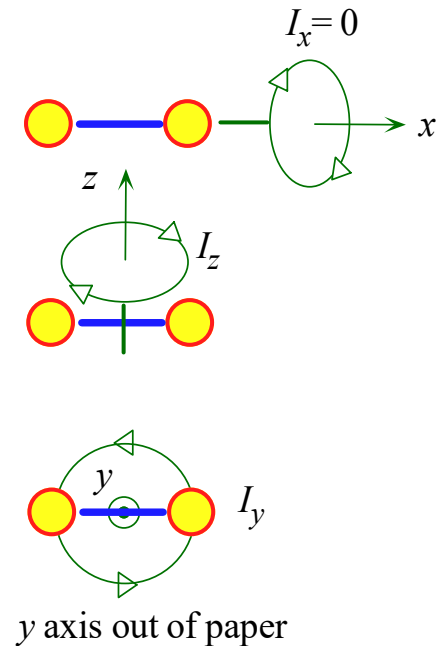
平动

TRANSLATIONAL MOTION



转动

ROTATIONAL MOTION



$$C_m = \frac{5}{2} R$$

## Diatomic molecule (刚性双原子分子)

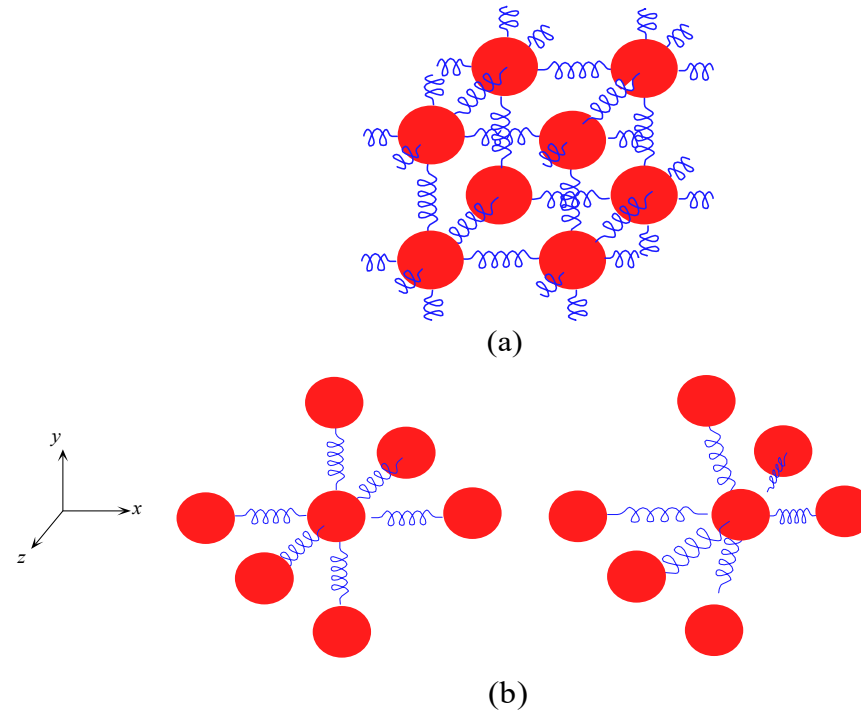
$$E = \frac{1}{2} m v_x^2 + \frac{1}{2} m v_y^2 + \frac{1}{2} m v_z^2 + \frac{1}{2} I_y \omega_y^2 + \frac{1}{2} I_z \omega_z^2$$

Each method by which a system can absorb energy is called a **degree of freedom**. (自由度)



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# Solid (固体)



(a) The ball-and-spring model of solids in which the springs represent the interatomic bonds. Each ball (atom) is linked to its neighbors by springs. Atomic vibrations in a solid involve 3 dimensions.

(b) An atom vibrating about its equilibrium position stretches and compresses its springs to the neighbors and has both kinetic and potential energy.

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{1}{2}K_x x_x^2 + \frac{1}{2}K_y x_y^2 + \frac{1}{2}K_z x_z^2$$

$$U = N_A 6 \left( \frac{1}{2} kT \right) = 3RT$$

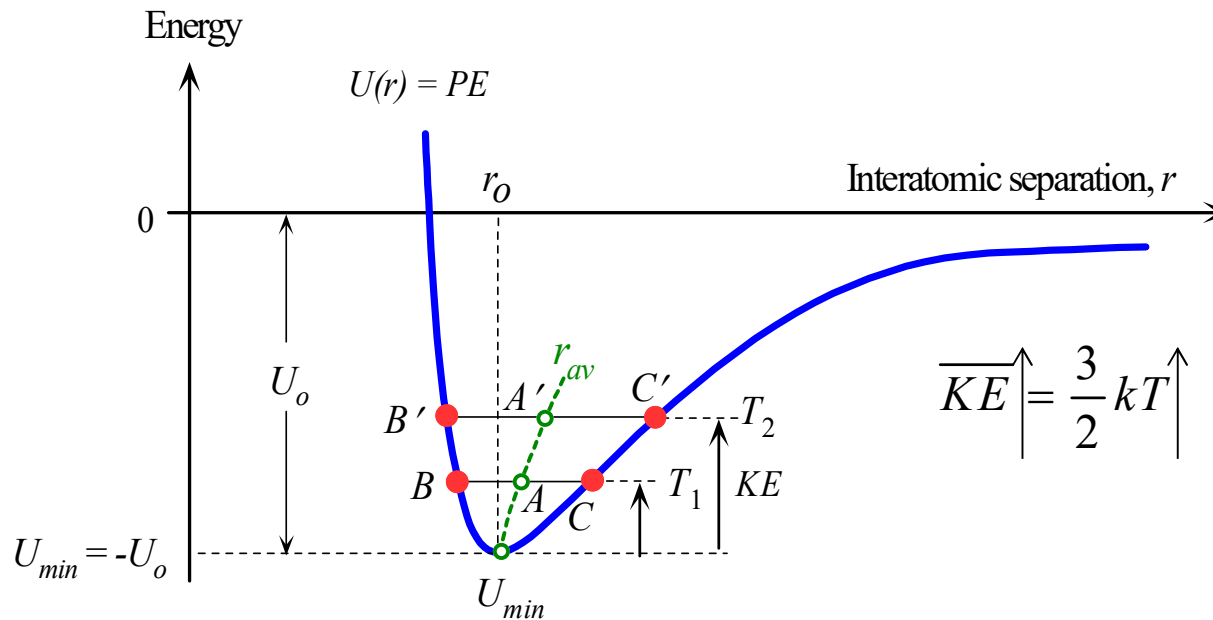
$$C_m = \frac{dU}{dT} = 3R = 25 Jk^{-1}mol^{-1}$$

Molar heat capacity: 摩尔热容



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## 1.4.2 Thermal expansion 热膨胀

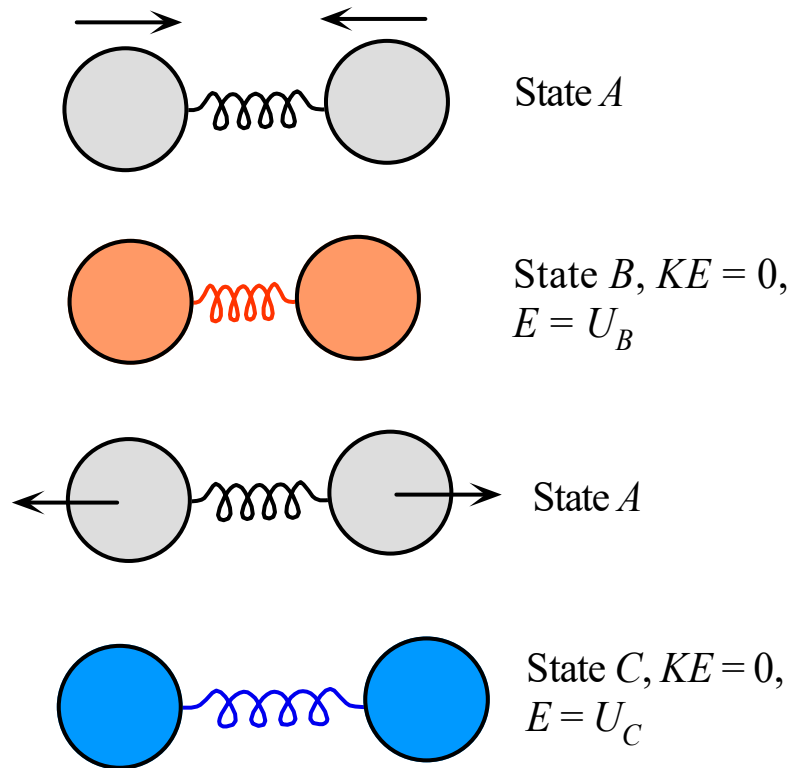


Nearly all materials expand as the temperature increases. This phenomenon is due to the **asymmetric** nature of the interatomic forces and the increase in the amplitude of atomic vibrations with temperature as expected from the kinetic molecular theory.

The potential energy  $PE$  curve has a minimum when the atoms in the solid attain the interatomic separation at  $r = r_0$ . Due to thermal energy, the atoms will be vibrating and will have vibrational kinetic energy. At  $T = T_1$ , the atoms will be vibrating in such a way that the bond will be stretched and compressed by an amount corresponding to the  $KE$  of the atoms. A pair of atoms will be vibrating between  $B$  and  $C$ . Their average separation will be at  $A$  and greater than  $r_0$ .

材料受热膨胀是由于原子间作用力的**不对称性**以及分子动力学理论所预计的原子的振动幅度随温度增加而增大。





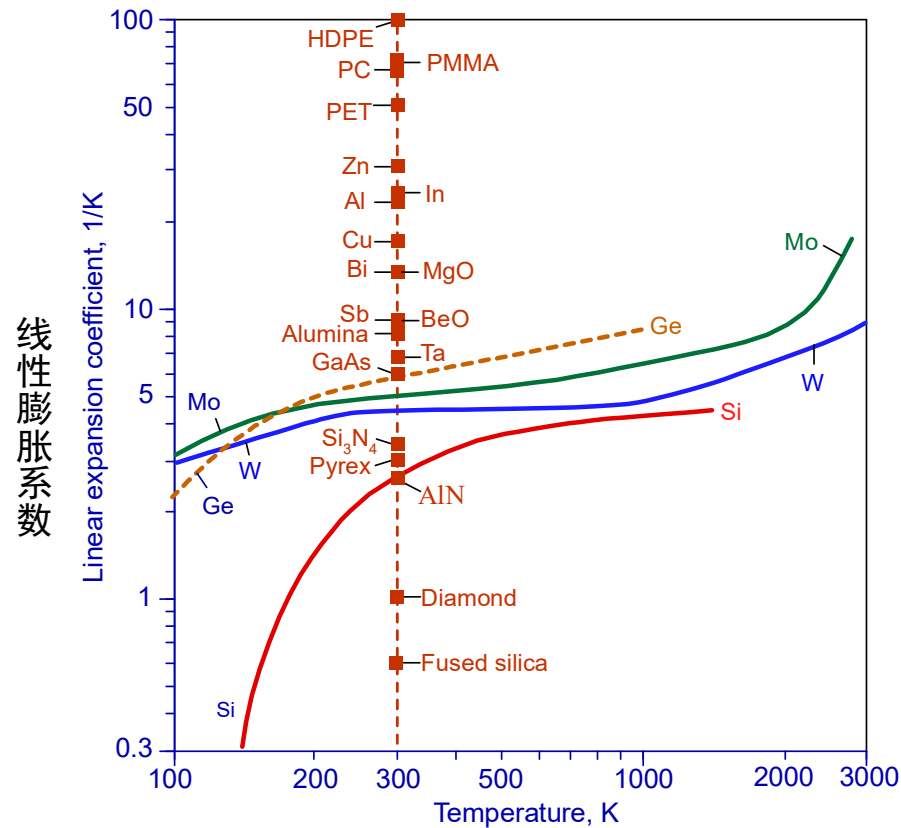
The PE curve  $U(r)$  is asymmetric. When the temperature increases, the average separation between the atoms also increases, which leads to the phenomenon of **thermal expansion**.

由于势能曲线不对称，当温度增加，平均间距距离增加，这就导致了**热膨胀**。

Vibrations of atoms in the solid. We consider, for simplicity a pair of atoms. Total energy is  $E = PE + KE$  and this is constant for a pair of vibrating atoms executing simple harmonic motion. At B and C,  $KE$  is zero (atoms are stationary and about to reverse direction of oscillation) and  $PE$  is maximum.







Dependence of the linear thermal expansion coefficient  $\lambda(K^{-1})$  on temperature  $T(K)$  on a log-log plot. HDPE, high density polyethylene; PMMA, Polymethylmethacrylate (acrylic); PC, polycarbonate; PET, polyethylene terephthalate (polyester); fused silica,  $SiO_2$ ; alumina,  $Al_2O_3$ .

SOURCE: Data extracted from various sources including G.A. Slack and S.F. Bartram, *J. Appl. Phys.*, **46**, 89, 1975.

$$\lambda = \frac{1}{L_0} \frac{\delta L}{\delta T}$$

Thermal expansion coefficient  $\lambda$ , 热膨胀系数。  
The fractional change in length per unit temperature

linear

$$L = L_0 [1 + (T - T_0) \lambda]$$





**VOLUME EXPANSION COEFFICIENT** Suppose that the volume of a solid body at temperature  $T_o$  is  $V_o$ . The volume expansion coefficient  $\alpha_V$  of a solid body characterizes the change in its volume from  $V_o$  to  $V$  due to a temperature change from  $T_o$  to  $T$  by

$$V = V_o[1 + \alpha_V(T - T_o)] \quad [1.20]$$

*Volume  
expansion*

Show that  $\alpha_V$  is given by

$$\alpha_V = 3\lambda \quad [1.21]$$

*Volume  
expansion  
coefficient*

Aluminum has a density of  $2.70 \text{ g cm}^{-3}$  at  $25^\circ\text{C}$ . Its thermal expansion coefficient is  $24 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ . Calculate the density of Al at  $350^\circ\text{C}$ .

### SOLUTION

Consider the solid body in the form of a rectangular parallelepiped with sides  $x_o$ ,  $y_o$ , and  $z_o$ . Then at  $T_o$ ,

$$V_o = x_o y_o z_o$$

and at  $T$ ,

$$\begin{aligned} V &= [x_o(1 + \lambda \Delta T)][y_o(1 + \lambda \Delta T)][z_o(1 + \lambda \Delta T)] \\ &= x_o y_o z_o (1 + \lambda \Delta T)^3 \end{aligned}$$

that is

$$V = x_o y_o z_o [1 + 3\lambda \Delta T + 3\lambda^2 (\Delta T)^2 + \lambda^3 (\Delta T)^3]$$

We can now substitute for  $V$  from Equation 1.20, use  $V_o = x_o y_o z_o$ , and neglect the  $\lambda^2 (\Delta T)^2$  and  $\lambda^3 (\Delta T)^3$  terms compared with the  $\lambda \Delta T$  term ( $\lambda \ll 1$ ) to obtain,

$$V = V_o[1 + 3\lambda(T - T_o)] = V_o[1 + \alpha_V(T - T_o)]$$

Since density  $\rho$  is mass/volume, volume expansion leads to a density reduction. Thus,

$$\rho = \frac{\rho_o}{1 + \alpha_V(T - T_o)} \approx \rho_o[1 - \alpha_V(T - T_o)]$$

For Al, the density at  $350^\circ\text{C}$  is

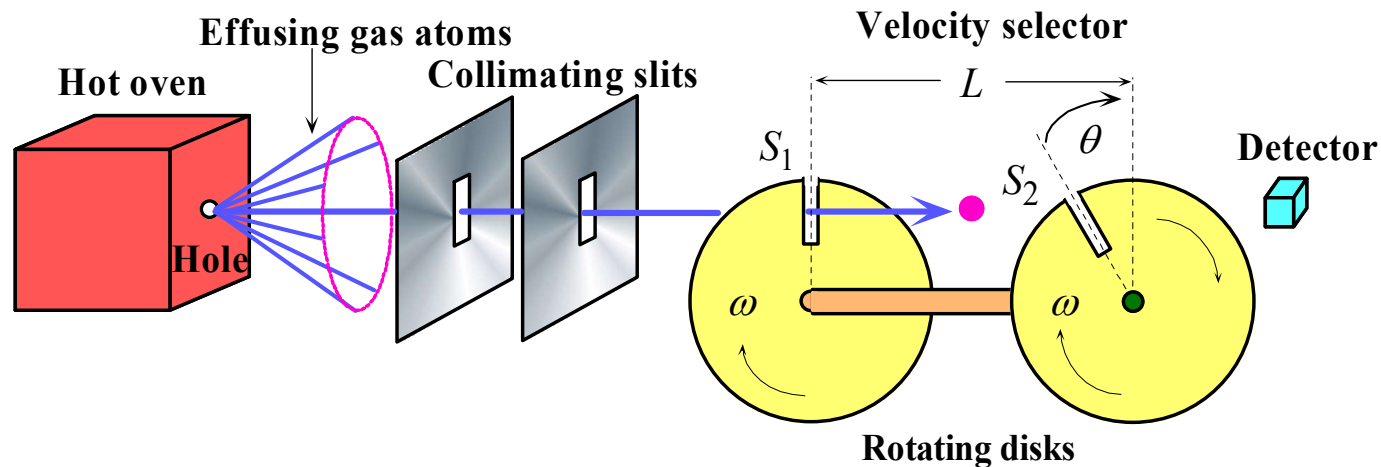
$$\rho = 2.70[1 - 3(24 \times 10^{-6})(350 - 25)] = 2.637 \text{ g cm}^{-3}$$



# 1.5 Molecular velocity and energy distribution

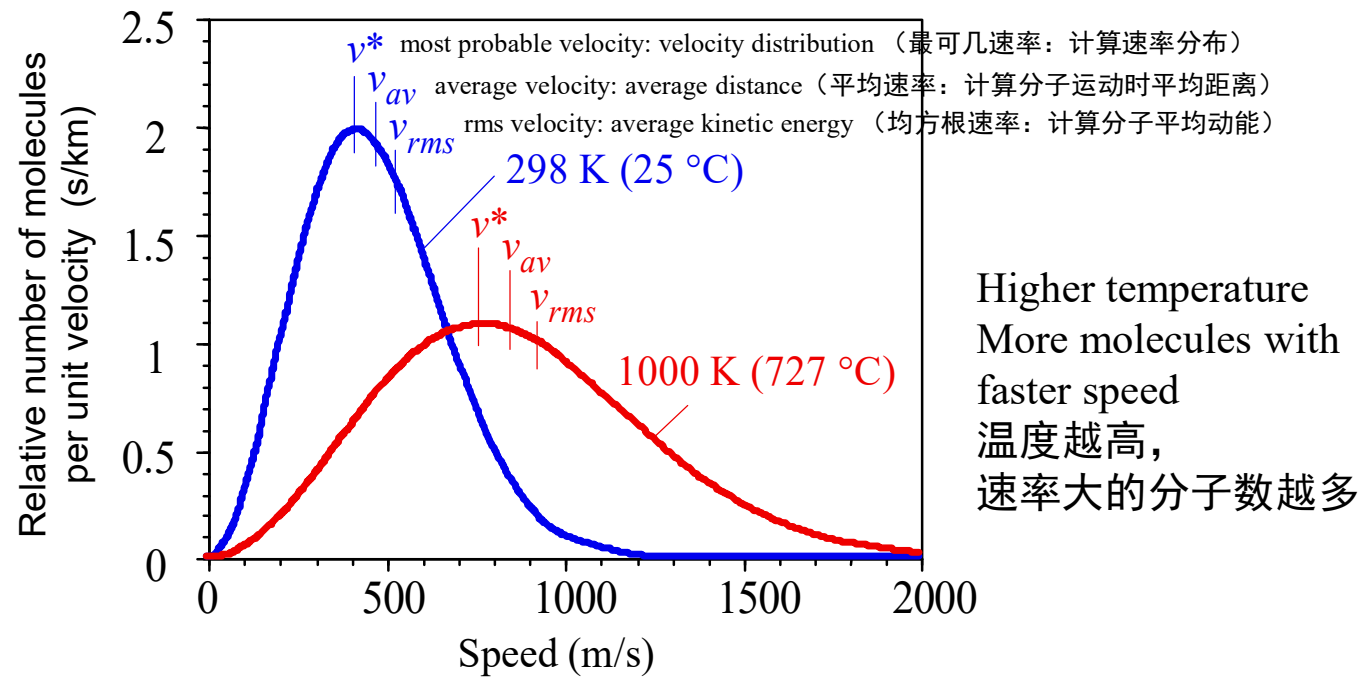
Due to random collisions between the molecules and the walls of the container and between the molecules themselves, the molecules do not all have the same velocity.  
由于分子间以及分子与容器壁的碰撞，分子不都具有相同的速率。

The number of molecular with speed from  $v$  to  $(v+dv)$



Schematic diagram of a Stern type experiment for determining the distribution of molecular speeds. 测定分子速度分布的斯特恩实验





Maxwell-Boltzmann distribution of molecular speeds in nitrogen gas at two temperatures. The ordinate is  $dN/(Ndv)$ , the fractional number of molecules per unit speed interval in  $(\text{km/s})^{-1}$

氮气在两个温度下的分子速率的麦克斯韦-玻尔兹曼分布，坐标为  $dN/(Ndv)$  和单位速率间隔内的分子数比例

$$n_v = 4\pi N \left( \frac{m}{2\pi kT} \right)^{3/2} v^2 \exp \left( -\frac{mv^2}{2kT} \right)$$

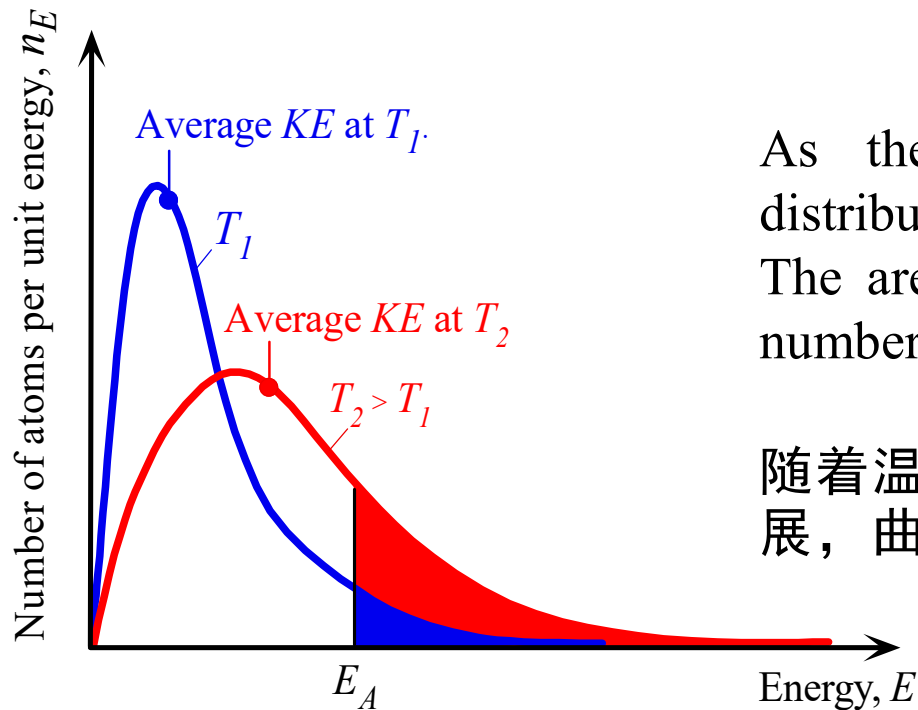
Maxwell-boltzmann distribution function  
麦克斯韦-玻耳兹曼分布



# Energy distribution of molecules

$$n_v = 4\pi N \left( \frac{m}{2\pi kT} \right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right)$$

$$n_e dE = n_v dv \quad \Rightarrow \quad n_E = \frac{2}{\pi^{1/2}} N \left( \frac{1}{kT} \right)^{3/2} E^{1/2} \exp\left(-\frac{E}{kT}\right)$$



As the temperature increases, the distribution extends to higher energies. The area under the curve is the total number of molecules.

随着温度增加，分布向高能方向扩展，曲线下的面积对应于分子总数。



**MEAN AND RMS SPEEDS OF MOLECULES** Given the Maxwell-Boltzmann distribution law for the velocities of molecules in a gas, derive expressions for the mean speed ( $v_{av}$ ), most probable speed ( $v^*$ ), and rms velocity ( $v_{rms}$ ) of the molecules and calculate the corresponding values for a gas of noninteracting electrons.

**SOLUTION**

The number of molecules with speeds in the range  $v$  to  $(v + dv)$  is

$$dN = n_v dv = 4\pi N \left( \frac{m}{2\pi kT} \right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right) dv$$

By definition, then, the mean speed is given by

$$v_{av} = \frac{\int v dN}{\int dN} = \frac{\int v n_v dv}{\int n_v dv} = \sqrt{\frac{8kT}{\pi m}}$$

where the integration is over all speeds ( $v = 0$  to  $\infty$ ). The mean square velocity is given by

$$\overline{v^2} = \frac{\int v^2 dN}{\int dN} = \frac{\int v^2 n_v dv}{\int n_v dv} = \frac{3kT}{m}$$

*Mean speed*

so the rms velocity is

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

where  $n_v/N$  is the probability per unit speed that a molecule has a speed in the range  $v$  to  $(v + dv)$ . Differentiating  $n_v$  with respect to  $v$  and setting this to zero,  $dn_v/dv = 0$ , gives the position of the peak of  $n_v$  versus  $v$ , and thus the most probable speed  $v^*$ ,

$$v^* = \left[ \frac{2kT}{m} \right]^{1/2}$$

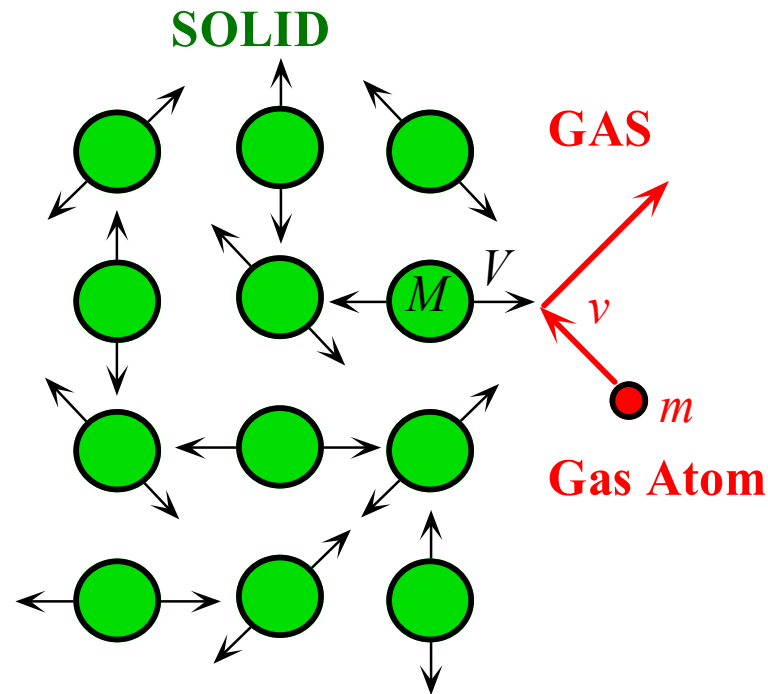
Substituting  $m = 9.1 \times 10^{-31}$  kg for electrons and using  $T = 300$  K, we find  $v^* = 95.3$  km s<sup>-1</sup>,  $v_{av} = 108$  km s<sup>-1</sup>, and  $v_{rms} = 117$  km s<sup>-1</sup>, all of which are close in value. We often use the term **thermal velocity** to describe the mean speed of particles due to their thermal random motion. Also, the integrations shown are not trivial and they involve substitution and integration by parts.

*Root mean  
square  
velocity*

*Most  
probable  
speed*



# 1.6 Heat, thermal fluctuations and noise

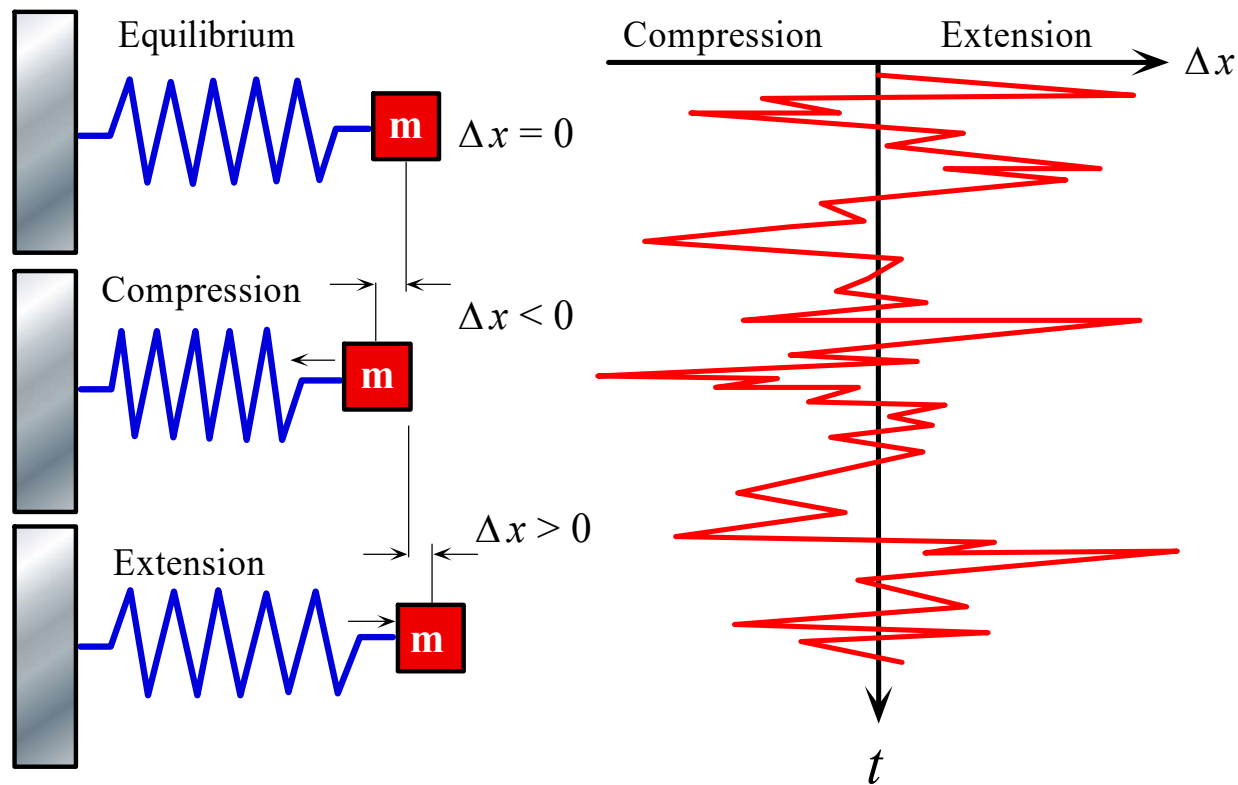


A solid in a He gas

Solid in equilibrium in air. During collisions between the gas and solid atoms, kinetic energy is exchanged.

The gas molecules will collide with the atoms on the surface of the solid and will thus exchange energy with those solid atoms. At the same temperature, over a long time, there will be **no net transfer of energy** from one to the other. **Thermal equilibrium (热平衡)**





由于空气分子随机撞击引起的弹簧质量块的伸缩

Fluctuations of a mass attached to a spring due to random bombardment by air molecules

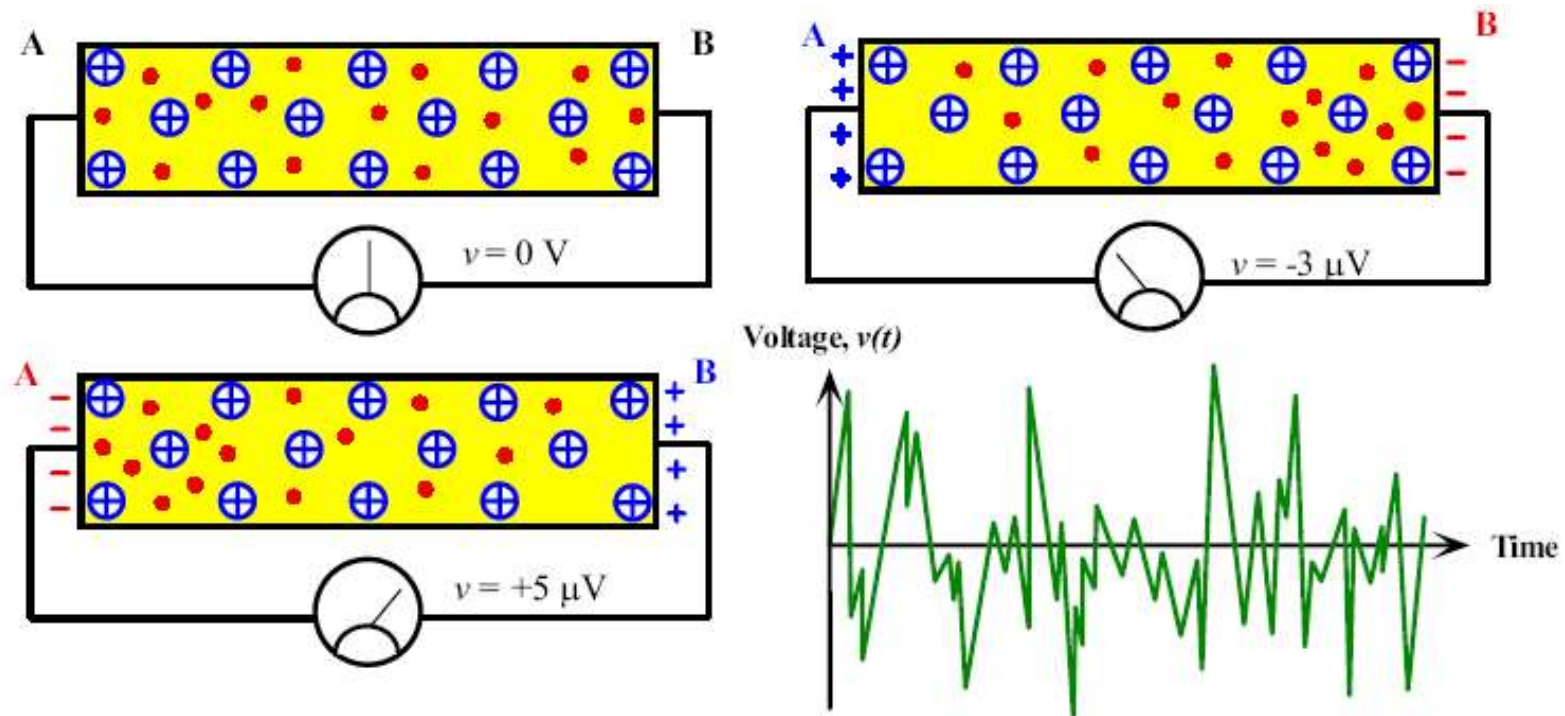
The energy stored as potential energy in the spring:  $PE(t) = \frac{1}{2} K (\Delta x)^2$

The average value of PE will be the same as KE,  $\overline{\frac{1}{2} K (\Delta x)^2} = \frac{1}{2} kT$



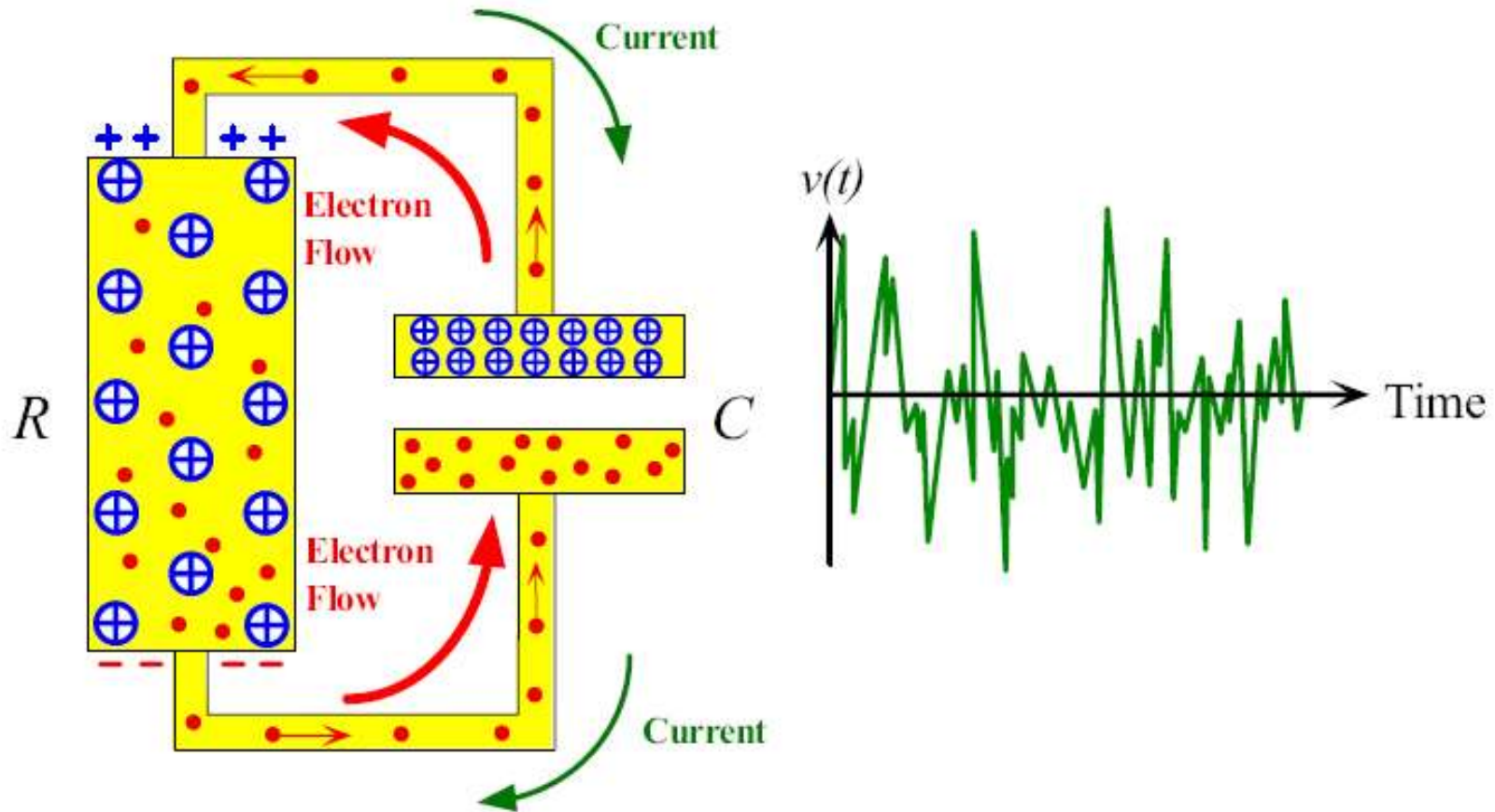


Random motion of conduction electrons in a conductor results in **electrical noise**  
导体中导电电子的随机运动引起**电噪声**



Because of the fluctuations in the electron concentration at any one instant, End A(B) of the conductor can become more negative with respect to end B(A)  
由于任意时刻电子密度的涨落，导体某端电位相对另一端有变化





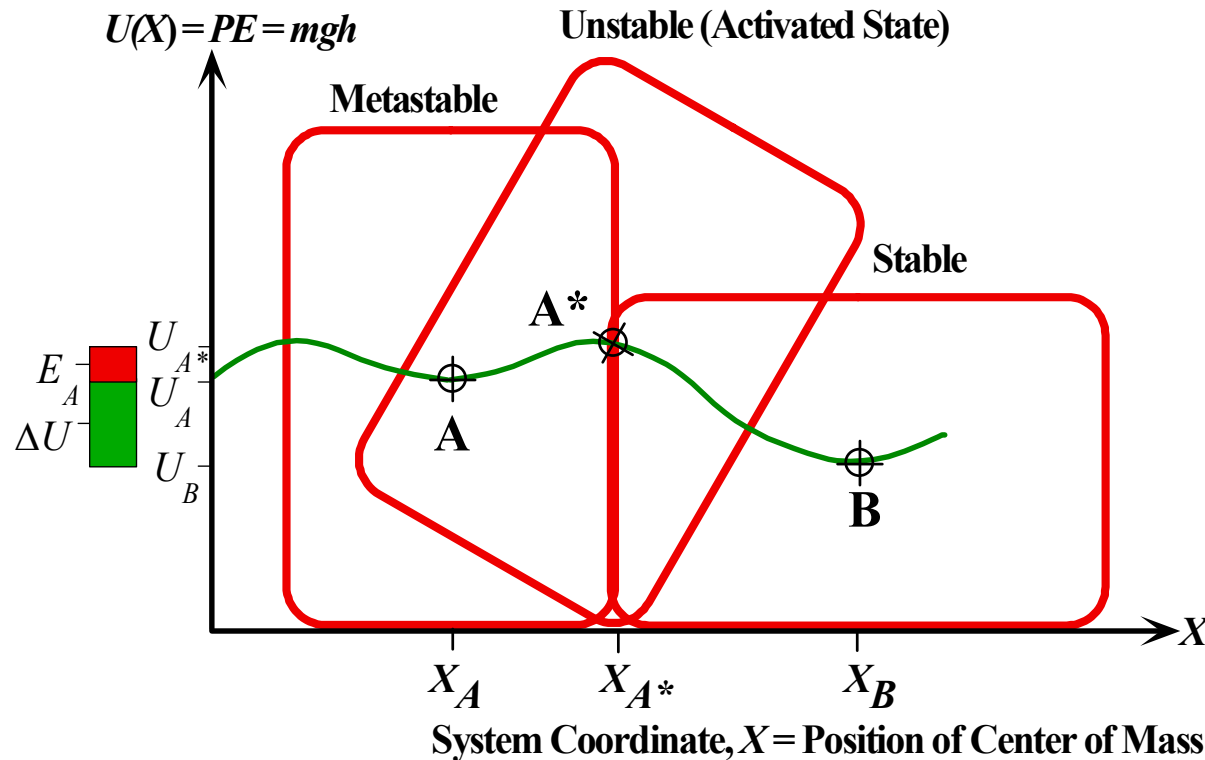
$$\overline{E(t)} = \frac{1}{2} C \overline{v(t)^2} = \frac{1}{2} kT$$

$$v_{rms} = \sqrt{4kTRB}$$

Johnson **resistor noise** equation  
 约翰逊**电阻噪声**方程



# 1.7 Thermally activated processes

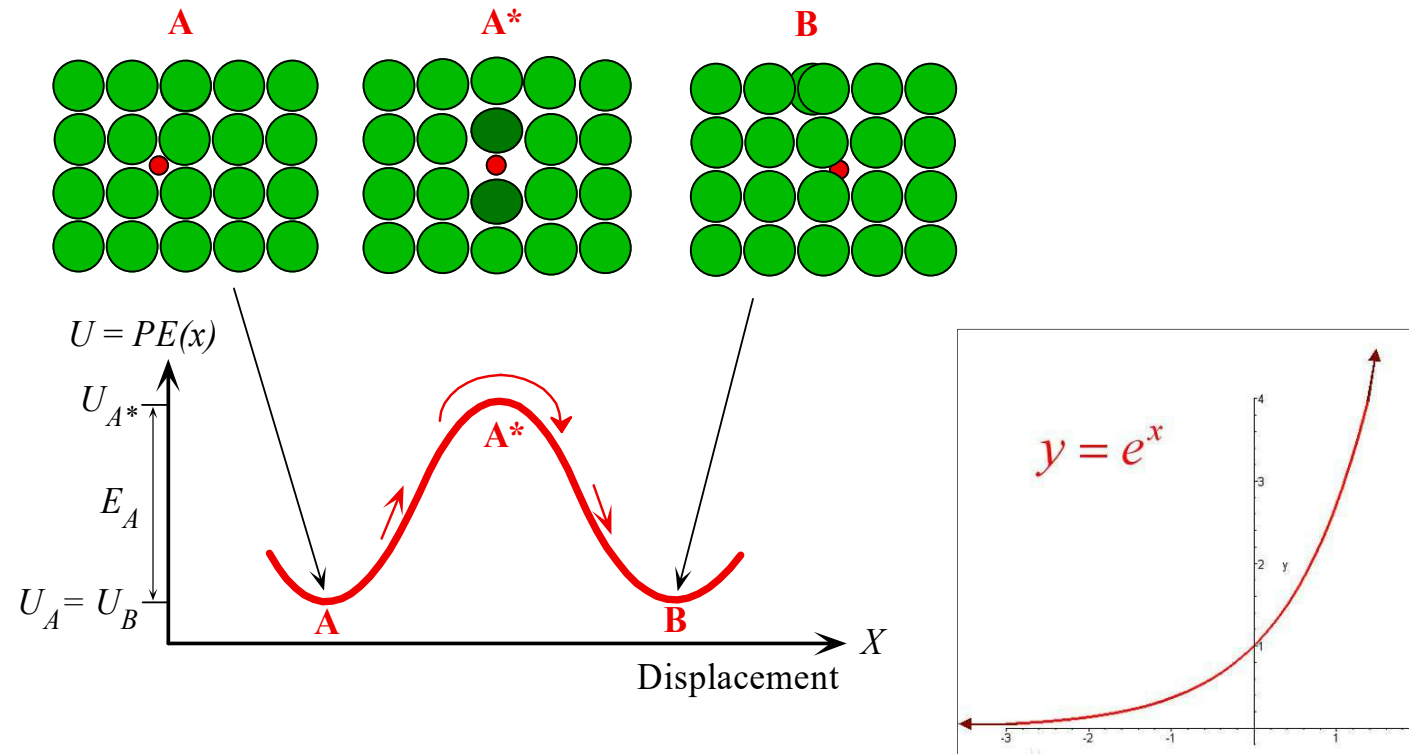


Tilting a failing cabinet from state A to its edge in state A\* requires an energy  $E_A$ . After reaching A\*, the cabinet spontaneously drops to the stable position B. PE of state B is lower than A and therefore state B is more stable than A.



The rate of jump, called the **diffusion (扩散)**

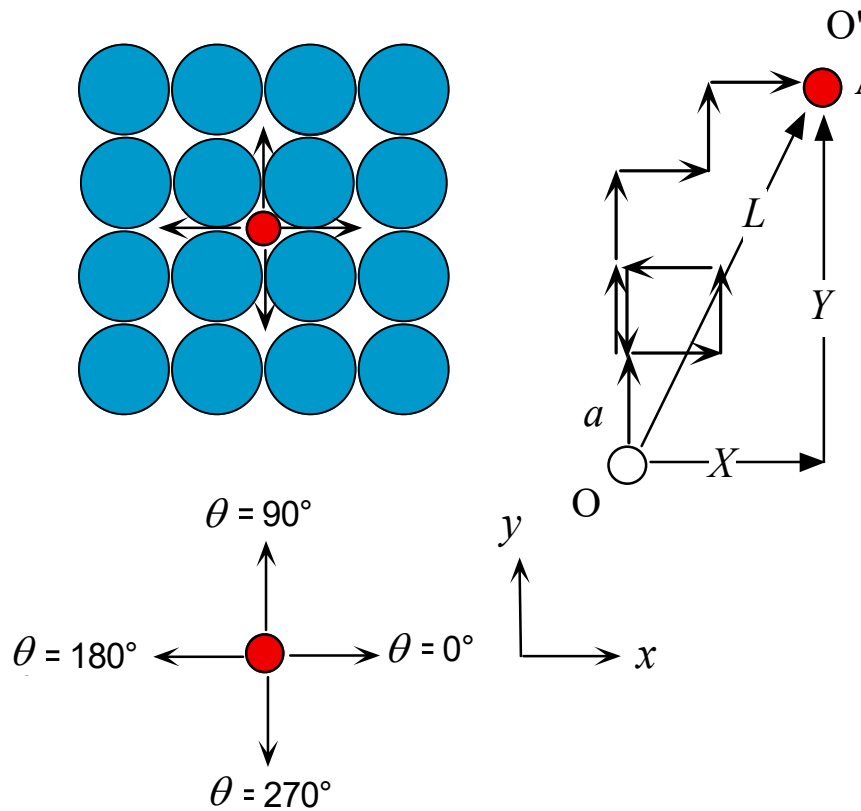
- 1) The number of times the atom tries to go over the potential barrier, which is the vibrational frequency  $V_o$ , in the AB direction
- 2) The probability that the atom has sufficient energy to overcome the PE barrier.



$$\begin{aligned} \text{Probability } (E > E_A) &= \frac{\text{Number of impurities with } E > E_A}{\text{Total number of impurities}} \\ &= \frac{\int_{E_A}^{\infty} n_E dE}{N} = A \exp\left(-\frac{E_A}{kT}\right) \end{aligned}$$

$$\begin{aligned} \vartheta &= (\text{Frequency of attempt along AB})(\text{Probability of } E > E_A) \\ &= A v_o \exp\left(-\frac{E_A}{kT}\right) \quad E_A = U_{A*} - U_A \end{aligned}$$





After  $N$  jumps

$$X^2 = a^2 \cos^2 \theta_1 + a^2 \cos^2 \theta_2 + \dots + a^2 \cos^2 \theta_N$$

$$= \frac{1}{2} a^2 N \quad \text{the same for } Y$$

Total square distance:  $L^2 = a^2 N$

Time:  $t = N/\mathcal{G}$   $N$ 次跃迁

$$L^2 = a^2 \mathcal{G} t = 2Dt$$

$$D = \frac{1}{2} a^2 \mathcal{G}$$

$$\mathcal{G} = A v_0 \exp\left(-\frac{E_A}{kT}\right)$$

$\mathcal{G}$ : Frequency of jump 跃迁频率

$$D = \frac{1}{2} a^2 v_0 A \exp\left(-\frac{E_A}{kT}\right)$$

$$= D_0 \exp\left(-\frac{E_A}{kT}\right)$$

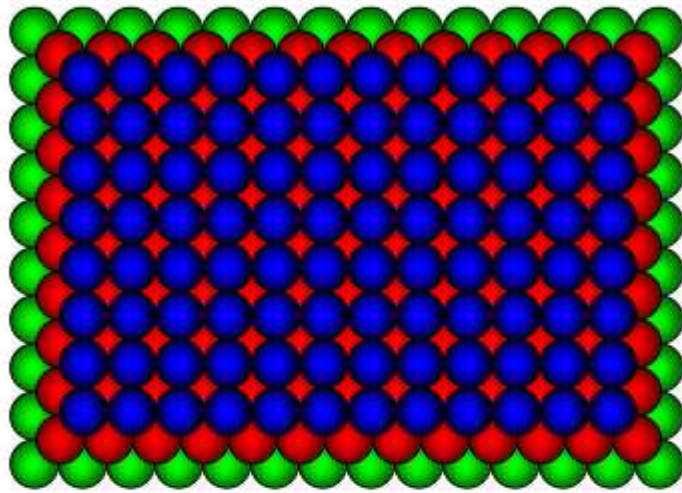
$D$ : Diffusion coefficient 扩散系数





# Crystalline and Amorphous

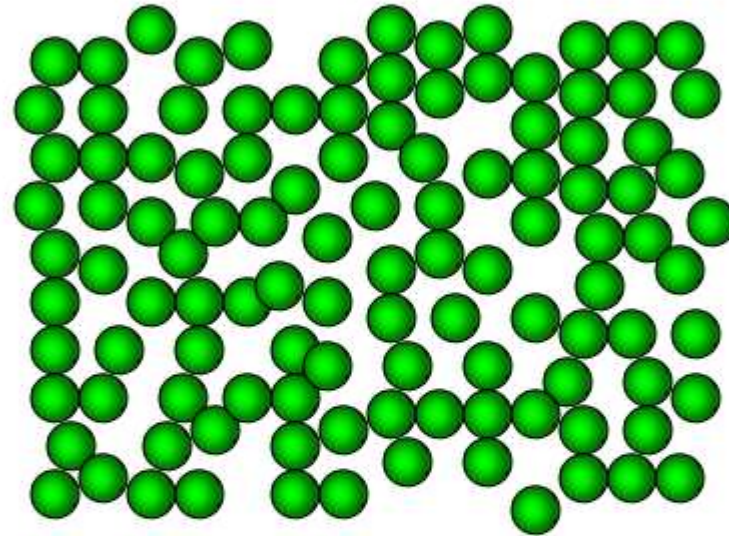
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**Grocery Store in the Morning**



**Cleaned Single Crystal Substrate  
Ready for High Quality Thin Film  
Growth**



**Grocery Store in the Evening**

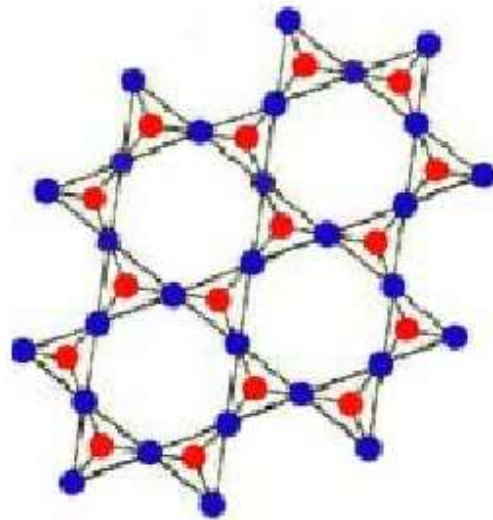


**Amorphous Substrate or Unclean  
substrate for Low Quality Film  
Growth**

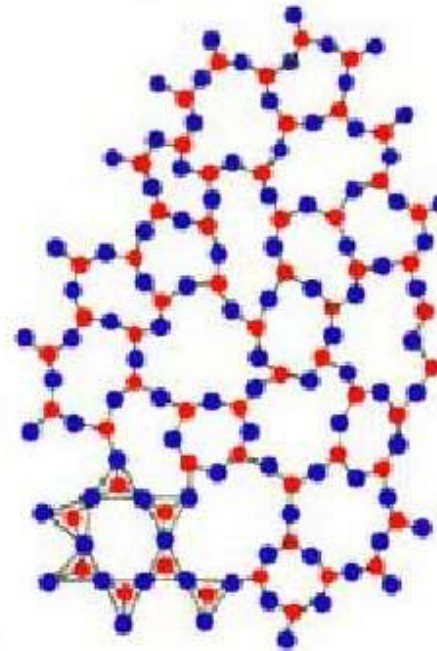


# Crystalline and amorphous SiO<sub>2</sub>

Crystalline SiO<sub>2</sub>  
(Quartz)



Amorphous SiO<sub>2</sub>  
(Glass)



• Si • O

Periodic structures vs. non-periodic structures

Crystal: Periodic structure repeats throughout the material.

Amorphous: no periodic structure at all.





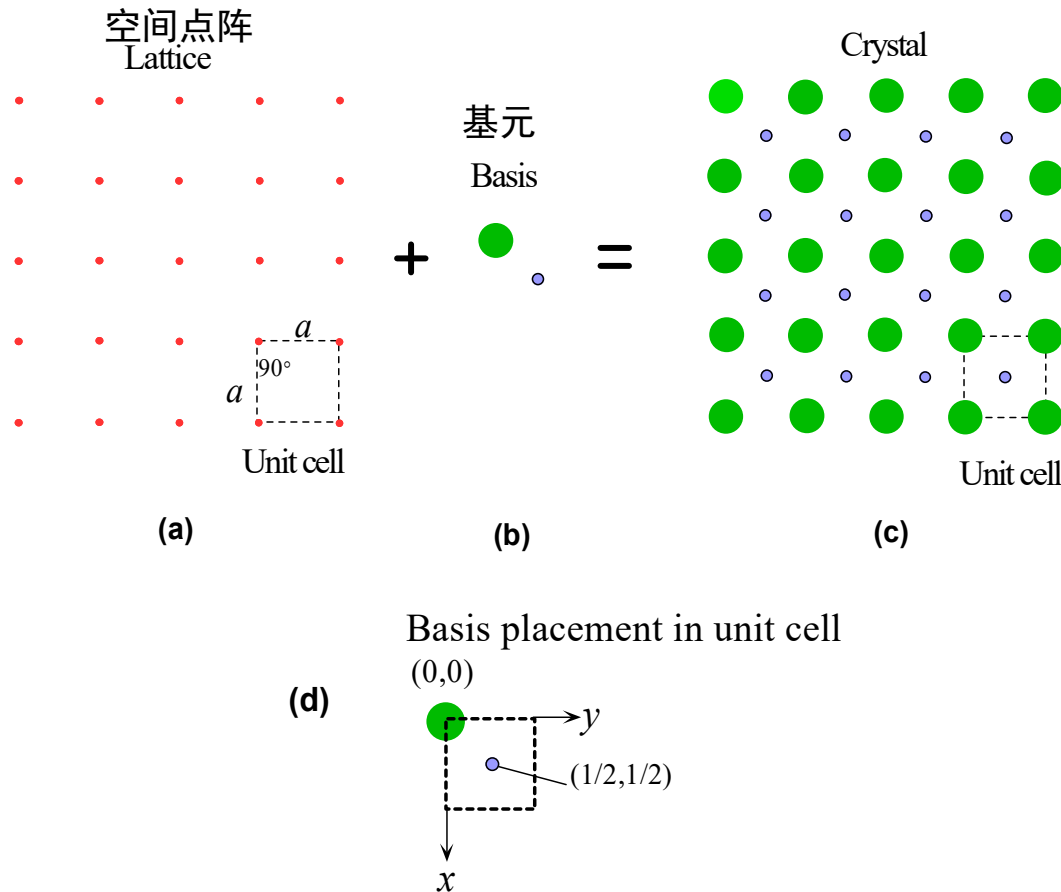
# Basic concepts: CRYSTAL STRUCTURE

---

- Atoms are considered as solid spheres with well defined radii
- **Lattice** - 3D array of points geometrically and repetitively distributed in space (atom center);
- Each lattice point has identical surroundings;
- The **unit cell** is the smallest structure that repeats itself by translation through the crystal. The most common types of unit cells are the face-centered cubic (FCC), the body-centered cubic (BCC) and the hexagonal close-packed (HCP).
  - smallest repetitive structure
  - defines crystal structure
- **Crystal structure = lattice + basis**



# Bravais lattices 布拉维格子



(a) A simple square lattice. The unit cell is a square with a side  $a$ . (b) Basis has two atoms. (c) Crystal = Lattice + Basis. The unit cell is a simple square with two atoms. (d) Placement of basis atoms in the crystal unit cell.

An infinite periodic array of geometric points in space defines a space lattice (空间点阵) of simply a lattice (点阵) .

The lattice point -->

An identical group of atoms (Basis, 基元)

Crystal = Lattice + Basis

$a$ : lattice parameter, 晶胞参数



## 1.8 Crystal Structures 晶体结构



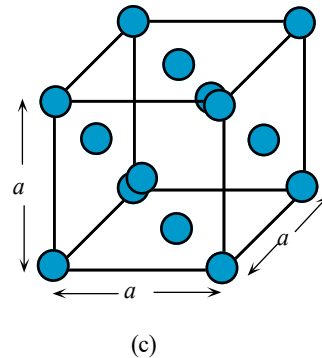
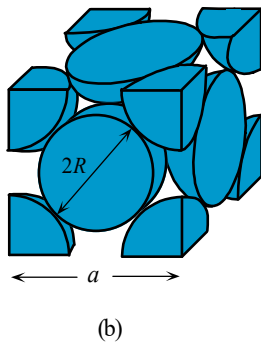
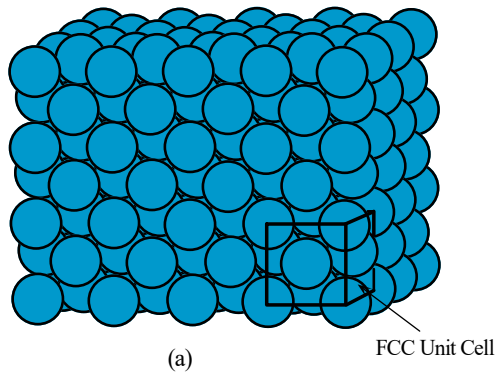
### Crystalline solid

the atoms bond with each other in a regular pattern to form a periodic collection of atoms in a solid.

Periodicity (周期性) ; long-range order (长程有序)



# Face-centered cubic (FCC) 面心立方



The unit cell of the copper crystal is cubic with Cu atoms at its **corner** and one Cu atom at the **center of each face**.

Lattice parameter  $a$  (晶胞参数 $a$ ): the length of the cubic unit.

For Cu:  $a = 0.362 \text{ nm}$

$$R = 0.128 \text{ nm}$$

$$\sqrt{2}a = 4R$$

How many atoms in the FCC unit cell?  
Coordination number?

$$8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4$$

$$C_n = 12$$

(a) The crystal structure of copper is Face Centered Cubic (FCC). The atoms are positioned at well defined sites arranged periodically and there is a long range order in the crystal.

(b) An FCC unit cell with closed packed spheres.

(c) Reduced sphere representation of the FCC unit cell. Examples: Ag, Al, Au, Ca, Cu,  $\gamma$ -Fe ( $>912^\circ\text{C}$ ), Ni, Pd, Pt, Rh

FCC structure: close-packed crystal structure (密堆积结构) The volume of the FCC unit cell is 74 percent full of atoms, which is the maximum packing possible with identical spheres.



## Metallic - fcc

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- Atomic packing factor: 原子堆积因子

$$APF = \frac{\text{Volume of atoms in unit cell}}{\text{Volume of unit cell}}$$

\*assume hard spheres

- Calculate the APF for the FCC crystal structure

$$APF = \frac{4 \times \frac{4}{3} \pi r^3}{a^3} \approx 74\%$$



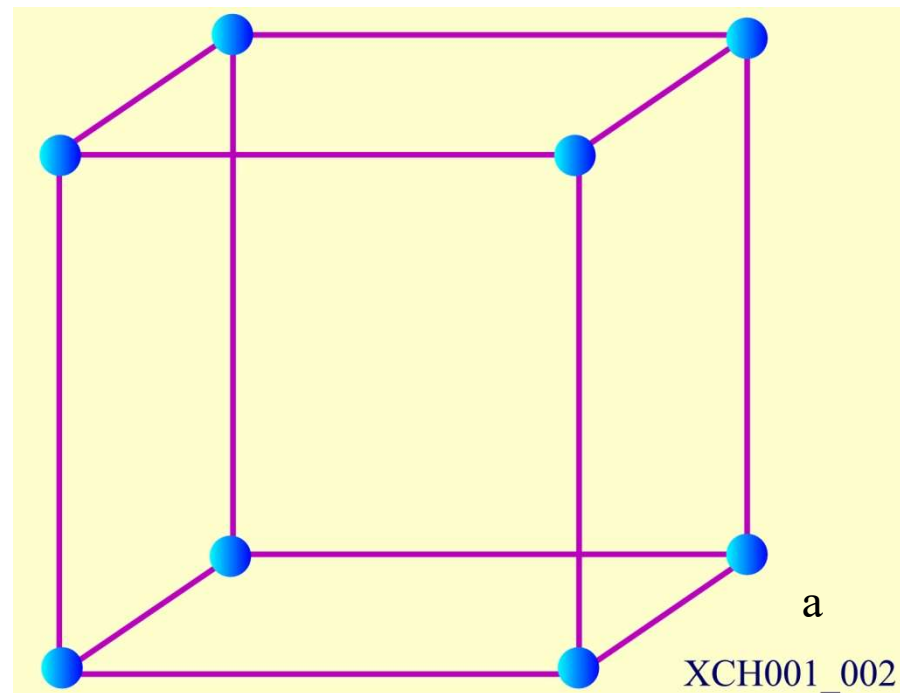


# Simple cubic 简单立方

Eight atoms at the corners. 原子球占据立方体的8个顶点；

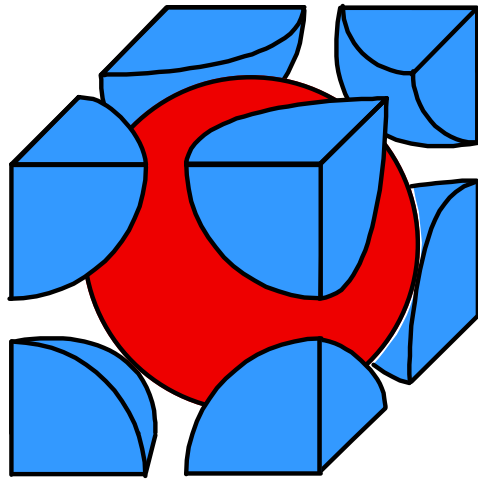
Coordination number ( $C_n$ : 配位数) = 6

Lattice parameter  $a$  (晶胞参数  $a$ )

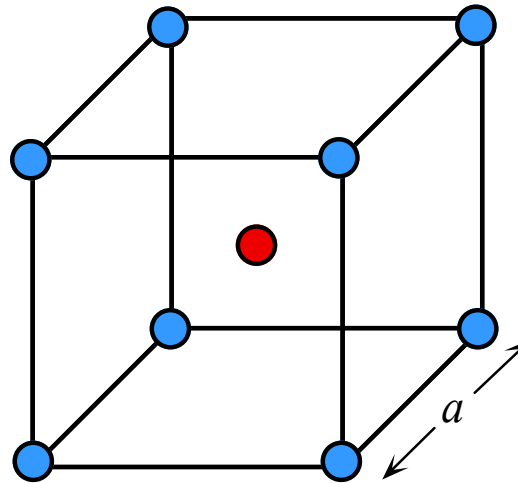




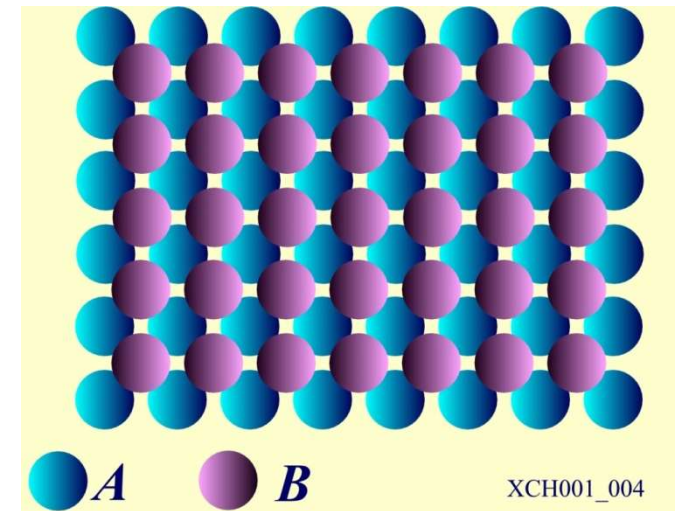
# Body-centered cubic (BCC) 体心立方



a



b



Examples: Alkali metals (Li, Na, K, Rb), Cr, Mo, W, Mn,  $\alpha$ -Fe ( $< 912^\circ\text{C}$ ),  $\beta$ -Ti ( $> 882^\circ\text{C}$ ).

Body centered cubic (BCC) crystal structure. (a) A BCC unit cell with closely packed hard spheres representing the Fe atoms. (b) A reduced-sphere unit cell.

How many atoms in a unit cell? 2

one atom at the center of the cell, eight atoms at the corners.

The volume of the BCC unit cell is 68 percent full of atoms.

$$a = ?R$$

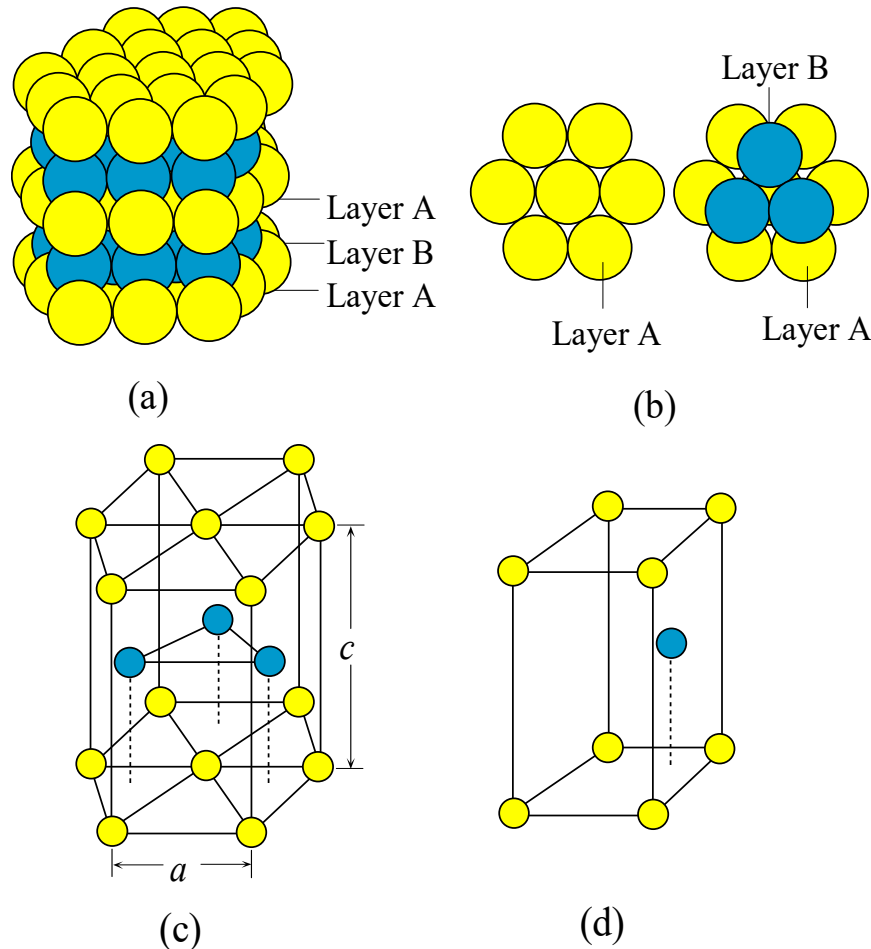
$$C_n = ?$$

$$a = \frac{4R}{\sqrt{3}}; \quad C_n = 8$$



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SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

# Hexagonal close-packed structure (HCP) 六方密堆积结构



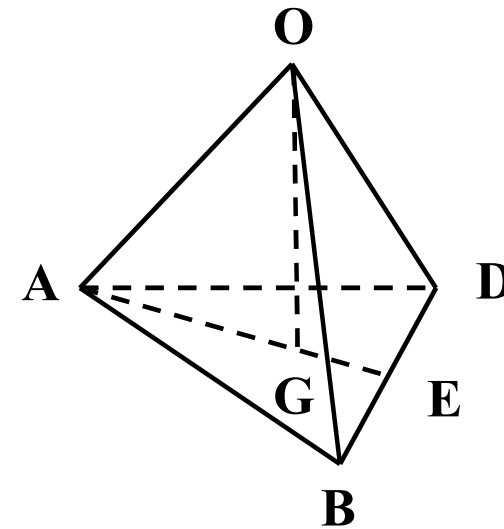
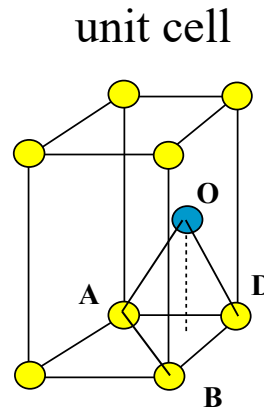
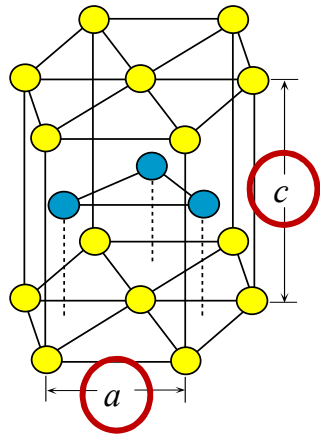
Examples: Be, Mg,  $\alpha$ -Ti ( $< 882^\circ\text{C}$ ), Cr, Co, Zn, Zr, Cd

The Hexagonal Close Packed (HCP) Crystal Structure.

- (a) The Hexagonal Close Packed (HCP) Structure. A collection of many Zn atoms. Color difference distinguishes layers (stacks).
- (b) The stacking sequence of closely packed layers is ABAB
- (c) A unit cell with reduced spheres
- (d) The smallest unit cell with reduced spheres.  $C_n = ?$  12

**Hexagonal close-packed structure(HCP)六方密堆积结构**, The atomic packing density in the HCP crystal structure is 74 percent, same as FCC structure.





1) Diameter

$$a = 2R$$

2) Relation between  $c$  and  $a$

$$AE = a \sin 60^\circ = \frac{\sqrt{3}}{2}a \quad AG = \frac{2}{3}AE = \frac{\sqrt{3}}{3}a$$

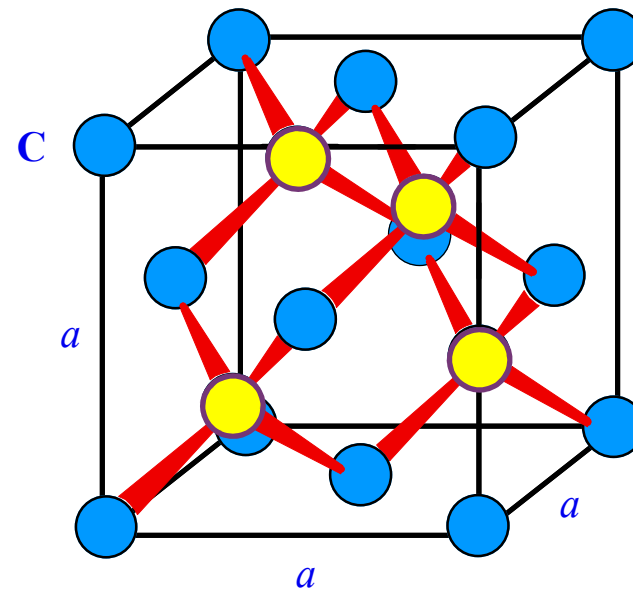
$$OG^2 + AG^2 = OA^2 \Rightarrow c = 2OG = \frac{2}{3}\sqrt{6}a$$

3) Atomic packing factor (APF)

$$APF = \frac{2 \frac{4}{3} \pi \left( \frac{a}{2} \right)^3}{\frac{\sqrt{3}}{2} a^2 c} = \frac{\sqrt{2}}{6} \pi$$



# Diamond cubic crystal structure 金刚石立方晶体结构



The diamond unit cell is cubic. The cell has eight atoms. Grey Sn ( $\alpha$ -Sn) and the elemental semiconductors Ge and Si have this crystal structure.

$$a = \frac{8R}{\sqrt{3}}; \quad C_n = 4$$

**Diamond cubic crystal structure 金刚石立方晶体结构.**

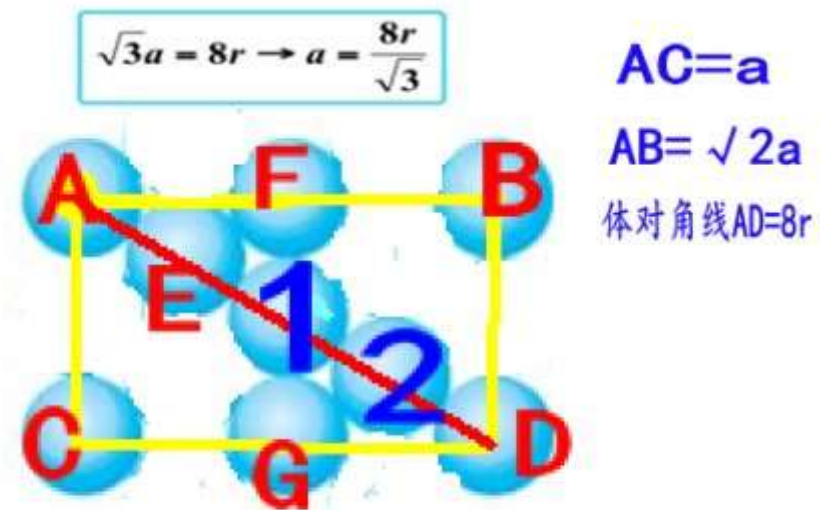
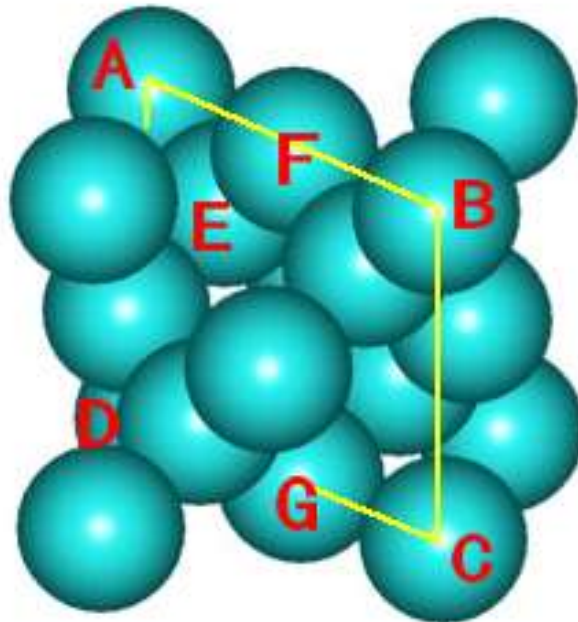
Eight atoms at the corner, six atoms at the center of each face, four atoms in the cell.



## Diamond cubic crystal structure 金刚石立方晶体结构.

Eight atoms at the corner, six atoms at the center of each face, four atoms in the cell.

$$a = \frac{8R}{\sqrt{3}}$$



1、2是虚拟的两个球



# Silicon

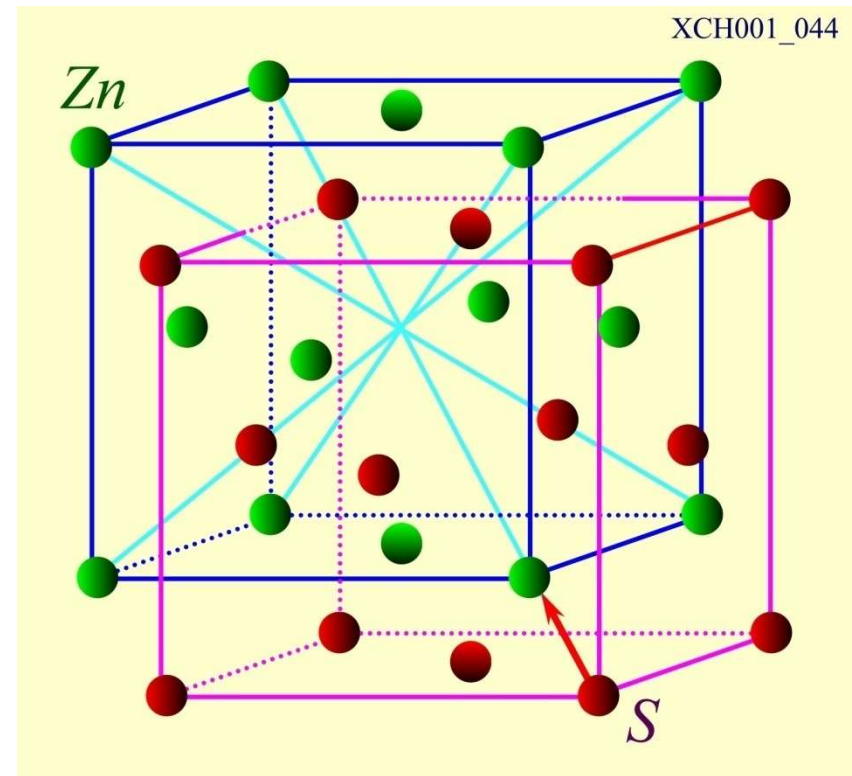
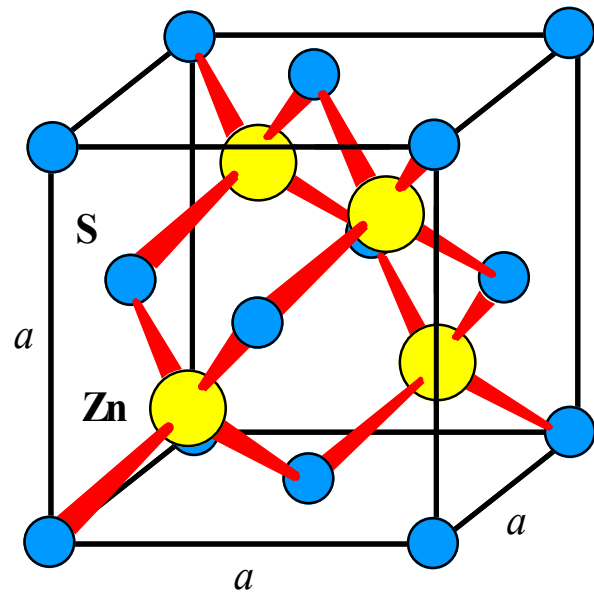


A silicon ingot is a single crystal of Si.





# Zinc blende structure 闪锌矿结构

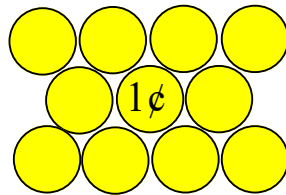


The Zinc blende (ZnS) cubic crystal structure. Many important compound crystals have the zinc blende structure. Examples: AlAs, GaAs, GaP, GaSb, InAs, InP, InSb, ZnS, ZnTe.



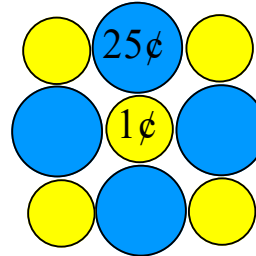
# Ionic solids 离子型晶体

Ratio of radii = 1

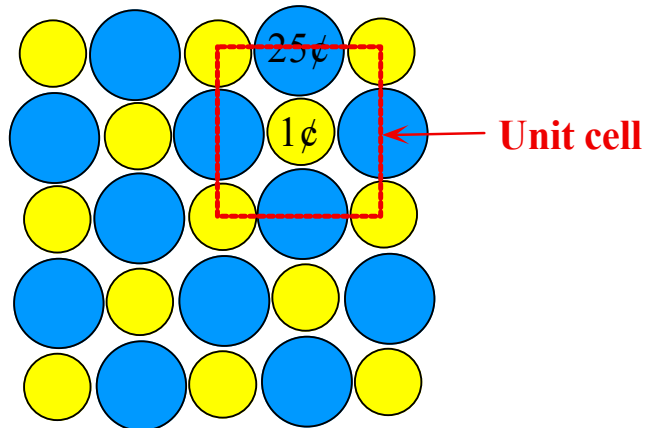


Nearest neighbors = 6

Ratio of radii = 0.75



Nearest neighbors = 4

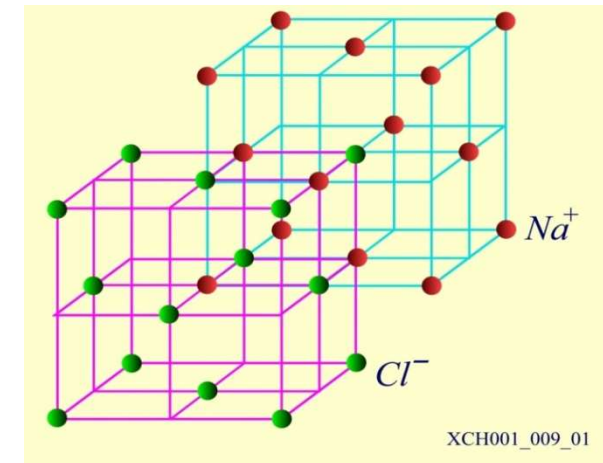
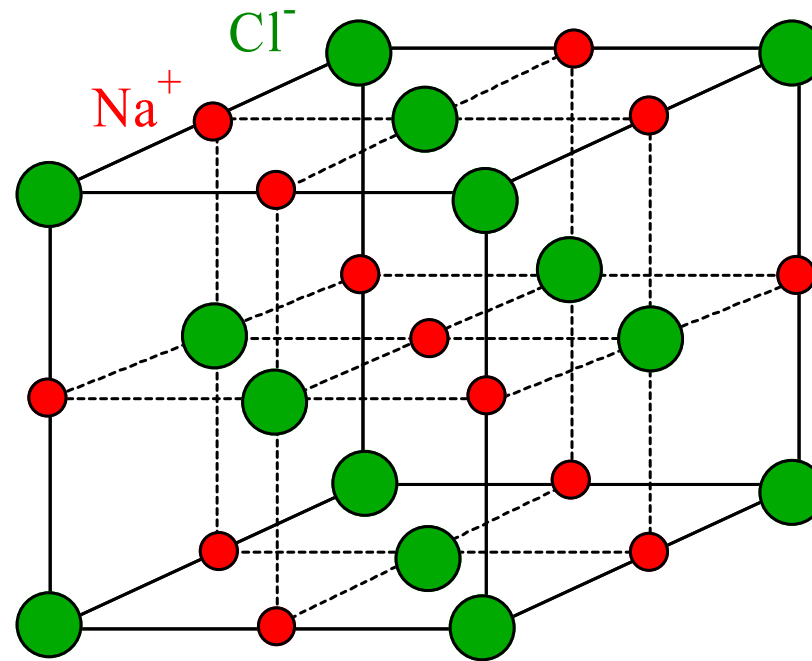


A two-dimensional crystal of pennies and quarters

Packing of coins on a table top to build a two dimensional crystal.



# NaCl unit cell 氯化钠晶胞



$$C_n = 6$$

A possible reduced sphere unit cell for the NaCl (rock salt) crystal.

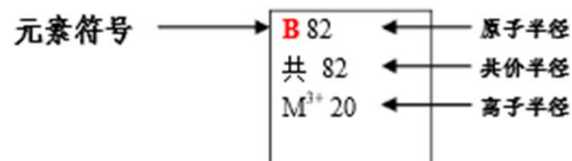
An alternative unit cell may have  $\text{Na}^+$  and  $\text{Cl}^-$  interchanged.

Examples: AgCl, CaO, CsF, LiF, LiCl, NaF, NaCl, KF, KCl, MgO

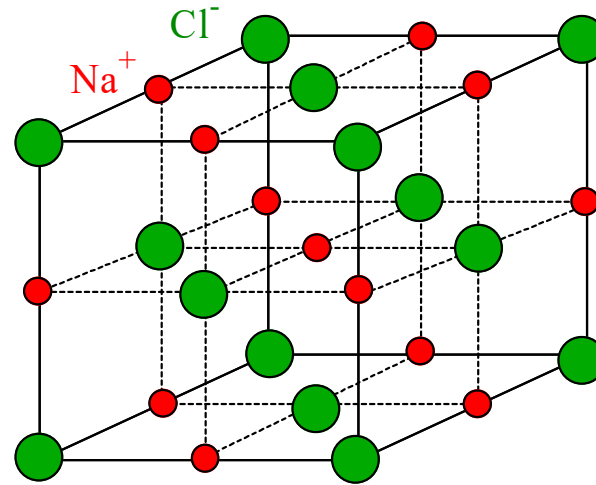


原子(离子)半径(pm)周期表

|                             |                             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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| H 32                        |                             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Li 123<br>M <sup>+</sup> 60 | Be 89<br>M <sup>2+</sup> 31 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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## NaCl unit cell

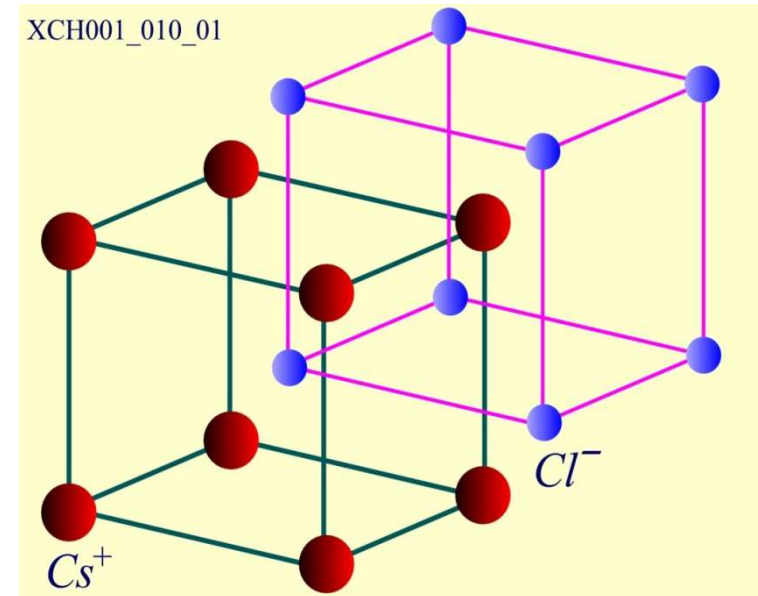
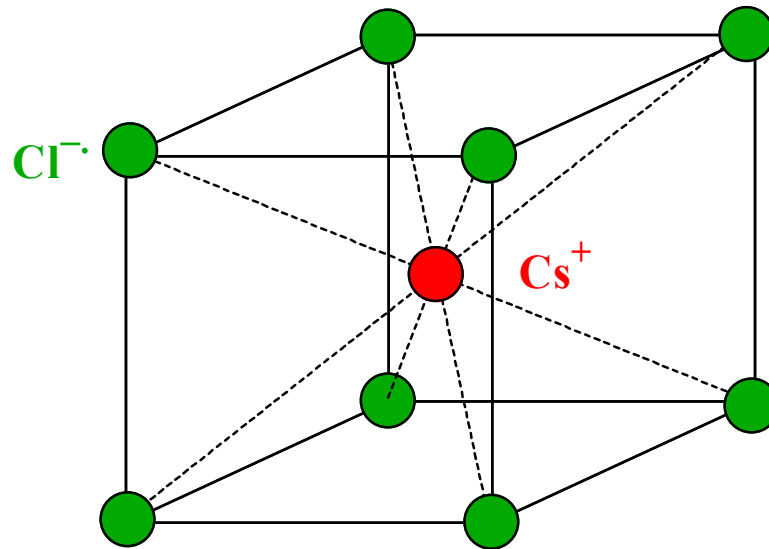


There are 4 Na<sup>+</sup> and 4 Cl<sup>-</sup> ions (共有4个Na离子和4个Cl离子)  
ion radius:  $r(\text{Na}^+) = 0.095 \text{ nm}$ ;  $r(\text{Cl}^-) = 0.181 \text{ nm}$

$$\text{APF} = \frac{4 \frac{4}{3} \pi (r)^3 + 4 \frac{4}{3} \pi (R)^3}{(2r + 2R)^3} = 0.67$$

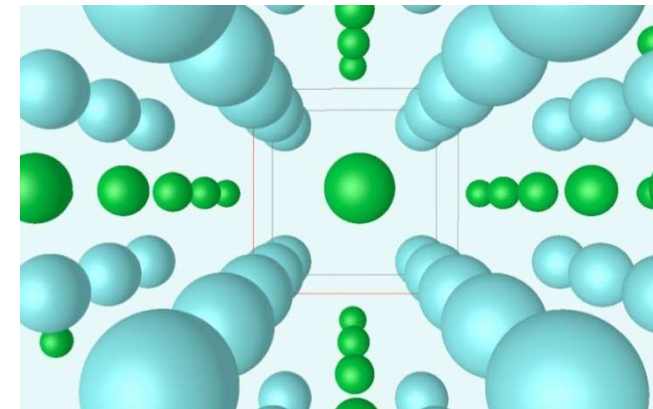


# CsCl structure



A possible reduced sphere unit cell for the CsCl crystal. An alternative unit cell may have  $\text{Cs}^+$  and  $\text{Cl}^-$  interchanged. Examples: CsCl, CsBr, CsI, TlCl, TlBr, TlI.

$$C_n = 8$$





**Table 1.3** Properties of some important crystal structures

| Crystal Structure | $a$ and $R$<br>( $R$ is the Radius of the Atom) | Coordination Number (CN) | Number of Atoms per Unit Cell | Atomic Packing Factor | Examples                                                                                                  |
|-------------------|-------------------------------------------------|--------------------------|-------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------|
| Simple cubic      | $a = 2R$                                        | 6                        | 1                             | 0.52                  | No metals (Except Po)                                                                                     |
| BCC               | $a = \frac{4R}{\sqrt{3}}$                       | 8                        | 2                             | 0.68                  | Many metals: $\alpha$ -Fe, Cr, Mo, W                                                                      |
| FCC               | $a = \frac{4R}{\sqrt{2}}$                       | 12                       | 4                             | 0.74                  | Many metals: Ag, Au, Cu, Pt                                                                               |
| HCP               | $a = 2R$<br>$c = 1.633a$                        | 12                       | 2                             | 0.74                  | Many metals: Co, Mg, Ti, Zn                                                                               |
| Diamond           | $a = \frac{8R}{\sqrt{3}}$                       | 4                        | 8                             | 0.34                  | Covalent solids:<br>Diamond, Ge, Si, $\alpha$ -Sn                                                         |
| Zinc blende       |                                                 | 4                        | 8                             | 0.34                  | Many covalent and ionic solids.<br>Many compound semiconductors.<br>ZnS, GaAs, GaSb, InAs, InSb           |
| NaCl              |                                                 | 6                        | 4 cations<br>4 anions         | 0.67<br>(NaCl)        | Ionic solids such as NaCl, AgCl, LiF, MgO, CaO<br>Ionic packing factor depends on relative sizes of ions. |
| CsCl              |                                                 | 8                        | 1 cation<br>1 anion           |                       | Ionic solids such as CsCl, CsBr, CsI                                                                      |

